Edward Y. Chang
Department of Electrical & Computer Engineering
Department of Computer Science (affiliated)
University of California
Santa Barbara, California 93106
Phone: (805) 893-2917
Email: echang@ece.ucsb.edu

EDUCATION

Ph.D.	Stanford University, Electrical Engineering (Advisor: Hector Garcia-Molina)	1995-99
M.S.	Stanford University, Computer Science	1992-94

AWARDS

NSF Career Award	2002
IBM Faculty Partnership Award	2000, 2001, 2002

EXPERIENCE

Associate Professor, University of California, Santa Barbara	2003-present
Co-Founder & CTO, VIMA Technologies, Santa Barbara	2001-present
Assistant Professor, University of California, Santa Barbara	1999-2003
Research Assistant, Stanford University	1995-99
Staff Engineer, Sun Microsystems, Mountain View, CA	1994-95

SELECTED PATENTS

"*Virtual IO: Preemptible Disk Access*," with University of California., 2002.
"*DPF: Dynamic Partial Function*," with VIMA Technologies Inc., 2002.
"*Perception-Based and Personalizable Image Retrieval*," with University of California, 2001.
"*Personalizable Interactive Digital TV*," with University of California, 2001.

PROFESSIONAL ACTIVITIES

Associated PC Chair for ACM MM 2002/2003
Panelist for National Science Foundtion 2002/2003
PC Member for IEEE MM 2002/2003, ICDE 2002, WISE 2002/2003, PAKDD 2002, WAIM 2002/2003, IS&T/SPIE 2002/2003/2004, ICCV 2003, IEEE CBAIVL 2000/2001.
Reviewer for ACM MM, SIGMOD, PODS, TODS, TOIS; IEEE ICDE, TKDE, TMM, etc.
Local Arrangement Chair for ACM SIGMOD 2001.

SELECTED INVITED TALKS

- "*Statistical Learning Methods for Emerging Database Applications*," Plenary Tutorial, 8^{th} International Conference on Database Systems for Advanced Applications, March 2003.
- "*Multimedia Information Retrieval Panelist*," International Conference on Multinedia, September 2002.
- "*Multimedia Information System Architecture*," Stanford Venture Seminar, May 2002.
- "*Multimedia Data Mining*," HP Labs., December 2001; January 2002.
- "*Dynamic Partial Function*," UC Berkeley Digital Library Seminar, February 2002.
- "*Measuring and Learning Perceptual Similarity*," UCLA, October 2001.
- "*Measuring and Learning Perceptual Similarity*," INRIA-Rocquencourt, September 2001.
- "*Learning Query Concepts via Intelligent Sampling*," IBM Almaden, September 2001.
- "*Learning Query Concepts via Intelligent Sampling*," IBM T.J. Watson, August 2001.
- "*Personalizable Interactive Digital VCR*," SONY Research, April 2001.
- "*On Managing Continuous Media Data*," Compaq Western Research Lab, September 2000.

TEN RECENT PUBLICATIONS

- *"Content-based Soft Annotation for Multimodal Image Retrieval Using Bayes Point Machines,"* E. Chang, K. Goh, G. Sychay, and G. Wu, IEEE Transactions on Circuits and Systems for Video Technology Special Issue on Conceptual and Dynamical Aspects of Multimedia Content Description , 2003 (to appear).
- *"MEGA: The Maximizing Expected Generalization Algorithm for Learning Complex Query Concepts,"* E. Chang and T. Li, in ACM Trans. on Information Systems, 2003 (to appear).
- *"The Discovery of A Percpeptual Distance Function for Measuring Image Similarity,"* B. Li, E. Chang, and Y. Wu, ACM Multimedia Journal Special Issue on Content-Based Image Retrieval, 2003 (to appear).
- *"Design and Implementation of Semi-Preemptible IO,"* Z. Dimitrijevic, R. Rangaswami, and E. Chang, in Proc. USENIX Conf. on File and Storage Technologies (FAST), April 2003.
- *"MEMS-based Disk Buffer for Streaming Media Servers,"* R. Rangaswami, Z. Dimitrijevic, E. Chang, and K. Schauser in Proc. of IEEE International Conference on Data Engineering, March 2003.
- *"DynDex: A Dynamic and Non-metric Space Indexer,"* K. Goh, B. Li, and E. Chang, in Proc. of the 9^{th} ACM International Conference on Multimedia, December 2002.
- *"Improving SVMs for Conducting Statistical Learning Under Extreme Constraints,"* G. Wu and E. Chang, UCSB Technical Report, November 2002 (submitted for publication).
- *"Approximate Similarity Queries in High-Dimensional Spaces,"* C. Li, E. Chang, H. Garcia-Molina, and G. Wiederhold, IEEE Transactions on Knowledge and Data Engineering, Volume 14, Number 4, pp.792-808, July/August 2002.
- *"Answering Queries with Useful Bindings,"* C. Li and E. Chang, ACM Transaction on Databases, Volume 2, Number 3, pp.313-343, September 2001.
- *"Support Vector Machine Active Learning For Image Retrieval,"* S. Tong and E. Chang, in Proc. of the 9^{th} ACM International Conference on Multimedia, pp.107-118, Ottawa, October 2001.

INVITED PAPERS

- *"Statistical Learning for Effective Visual Image Retrieval,"* E. Chang, B. Li, and G. Wu, IEEE International Conference on Image Processing, Barcelona, September 2003.
- *"On Learning Perceptual Distance Functions for Image Retrieval,"* B. Li and E. Chang, in Proc. of IEEE International Conference on Acoustics, Speech and Signal Processing, Orlando, May 2002.
- *"High-dimensional Visual Data Indexing,"* K. Goh and E. Chang, to appear in Proc. of the 6^{th} International Conference on Visual Databases, Australia, May 2002.
- *"Learning and Measuring Perceptual Similarity,"* E. Chang, in Proc. of the 1^{st} NSF/INRIA/Berkeley/IBM Workshop on Content-based Multimedia Information Retrieval, Paris, September, 2001.

COLLABORATORS & GRADUATE STUDENTS

Hector Garci-Molina (graduate advisor), Chen Li (collaborator), Simon Tong (collaborator), James Wang (collaborator), Gio Wielderhold (collaborator), Zoran Dimitrijevic (Phd candidate, CS), Kingshy Goh (Phd candidate, ECE), Beitao Li (Phd candidate, ECE), Yan Meng (Phd student, ECE), Raju Rangaswami (Phd candidate, CS), Gerard Sychay (MS, CS), Gang Wu (Phd candidate, ECE), Yi Wu (Phd candidate, ECE), Huaxin You (Phd candidate, Statistics).